

Paramount Diagnostic Lab

Test Details

METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Immunoassay for the in vitro quantitative determination of the sum of human chorionic gonadotropin (hCG) plus the hCG beta-subunit in human serum and plasma.¹ This assay is intended for the early detection of pregnancy.

Special Instructions

Values obtained with different assay methods should not be used interchangeably in serial testing. It is recommended that only one assay method be used consistently to monitor each patient's course of therapy. This procedure does not provide serial monitoring; it is intended for one-time use only. If serial monitoring is required, please use the serial monitoring number **480038** (/tests/480038/human-chorionic-gonadotropin-hcg-subunit-quantitative-serial-monitor) to order.

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

As with all tests containing monoclonal mouse antibodies, erroneous findings may be obtained from samples taken from patients who have been treated with monoclonal mouse antibodies or have received them for diagnostic purposes.¹ In rare cases, interference due to extremely high titers of antibodies to streptavidin and ruthenium can occur.¹ The test contains additives, which minimize these effects.

Custom Additional Information

Similarly to LH, FSH, and TSH, human chorionic gonadotropin (hCG) is a member of the glycoprotein family and consists of two subunits (α and β chains) that are associated to the intact hormone.¹⁻⁷ The α -chains in all four of these glycoprotein hormones are virtually identical, whereas the β -chains have greatly differing structures and are responsible for the respective specific hormonal functions.

hCG is produced in the placenta during pregnancy. In nonpregnant women, it can also be produced by tumors of the trophoblast, germ cell tumors with trophoblastic components, and some nontrophoblastic tumors.

Human chorionic gonadotropin consists of a number of isohormones with differing molecular size. The biological action of hCG serves to maintain the corpus luteum during pregnancy. It also influences steroid production. The serum of pregnant women contains mainly intact hCG.

Measurement of the hCG concentration permits the diagnosis of pregnancy just one week after conception. The determination of hCG in the first trimester of pregnancy is of particular importance. Elevated values here serve as an indication of chorionic carcinoma, hydatiform mole, or multiple pregnancy. Depressed values indicate threatening or missed abortion, ectopic pregnancy, gestosis, or intrauterine death.

Elevated hCG concentrations not associated with pregnancy are found in patients with other diseases, such as tumors of the germ cells, ovaries, bladder, pancreas, stomach, lungs, and liver.^{6,7}

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

If a red-top tube or plasma is used, transfer separated serum or plasma to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.

	Range (mIU/mL)
Male	0–3
Female	
nonpregnant	0–5
postmenopausal	0–8
pregnant	
Weeks Gestation	
3	6–71
4	10–750
5	217–7138
6	158–31,795
7	3697–163,563
8	32,065–149,571
9	63,803–151,410
10	46,509–186,977
12	27,832–210,612
Weeks of pregnancy is defined as completed weeks beginning with the start of the last menstruation phase.	

	Range (mIU/mL)
14	13,950–62,530
15	12,039–70,971
16	9040–56,451
17	8175–55,868
18	8099–58,176
Weeks of pregnancy is defined as completed weeks beginning with the start of the last menstruation phase.	

Storage Instructions

Room temperature

Causes for Rejection

Citrate plasma specimen; improper labeling

Footnotes

1. hCG+ β (Intact human chorionic gonadotropin + the β -subunit) on Elecsys 1010/2010 and Modular Analytics E170, 2016-08, V 17.0 English [package insert]. Indianapolis, Ind: Roche Diagnostics; 2016.

2. Thomas CM, Reijnders FJ, Segers MF, Doesburg WH, Rolland R. Human Choriogonadotropin (HCG): Comparisons between Determinations of Intact HCG, Free HCG β -Subunit, and “Total” HCG + β in Serum during the First Half of High-Risk Pregnancy. *Clin Chem*. 1990 Apr;36(4):651-655.

PubMed 1691055 (<http://www.ncbi.nlm.nih.gov/pubmed/1691055>)

3. Hoermann R, Berger P, Spoettl G, et al. Immunological Recognition and Clinical Significance of Nicked Human Chorionic Gonadotropin in

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4. Schwarz S, Berger P, Wick G. The Antigenic Surface of Human Chorionic Gonadotropin as Mapped by Murine Monoclonal Antibodies.

Endocrinology. 1986 Jan;118(1):189-197. PubMed 2416550

(<http://www.ncbi.nlm.nih.gov/pubmed/2416550>)

5. Runnebaum B, Rabe T. Gynäkologische Endokrinologie, Grundlagen, Physiologie, Pathologie, Prophylaxe, Diagnostik, Therapie. Berlin, Heidelberg, New York, London, Paris, Tokyo: *Springer Verlag*.

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6. Sturgeon CM, McAllister EJ. Analysis of hCG: clinical applications and assay requirements. *Ann Clin Biochem*. 1998 Jul;35(Pt 4):460-491. PubMed

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7. Marcillac I, Troalen F, Bidart JM, et al. Free Human Chorionic Gonadotropin β Subunit in Gonadal and Nongonadal Neoplasms. *Cancer Res*. 1992 Jul 15;52(14):3901-3907. PubMed 1377600

(<http://www.ncbi.nlm.nih.gov/pubmed/1377600>)